Wastewater-Based Epidemiology for Emerging Pathogens in a Northern California Community Hospital, November 2024 to April 2025

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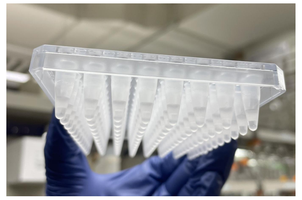
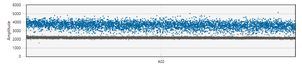
Background

- Wastewater-based epidemiology (WBE) has been effectively utilized to monitor respiratory viral circulation at the community level, its application in hospital settings remains largely unexplored.
- We used WBE to gain insights into emerging pathogen circulation within a hospital environment

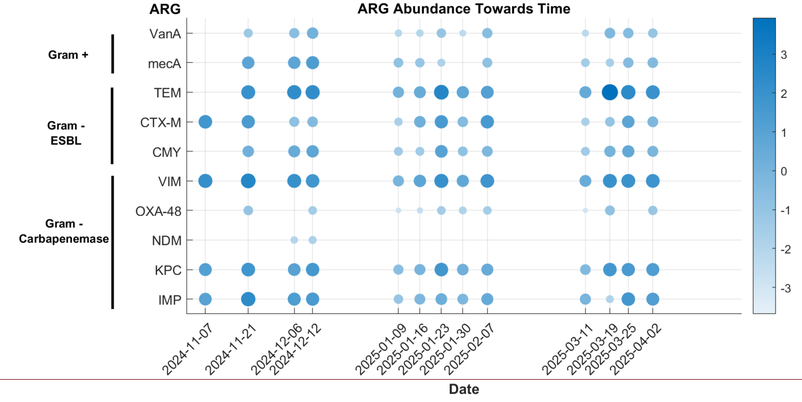
Methods

Weekly composite samples using autosampler → **Spined down solids and Nucleic Acids Extraction** → **Droplet digital PCR**

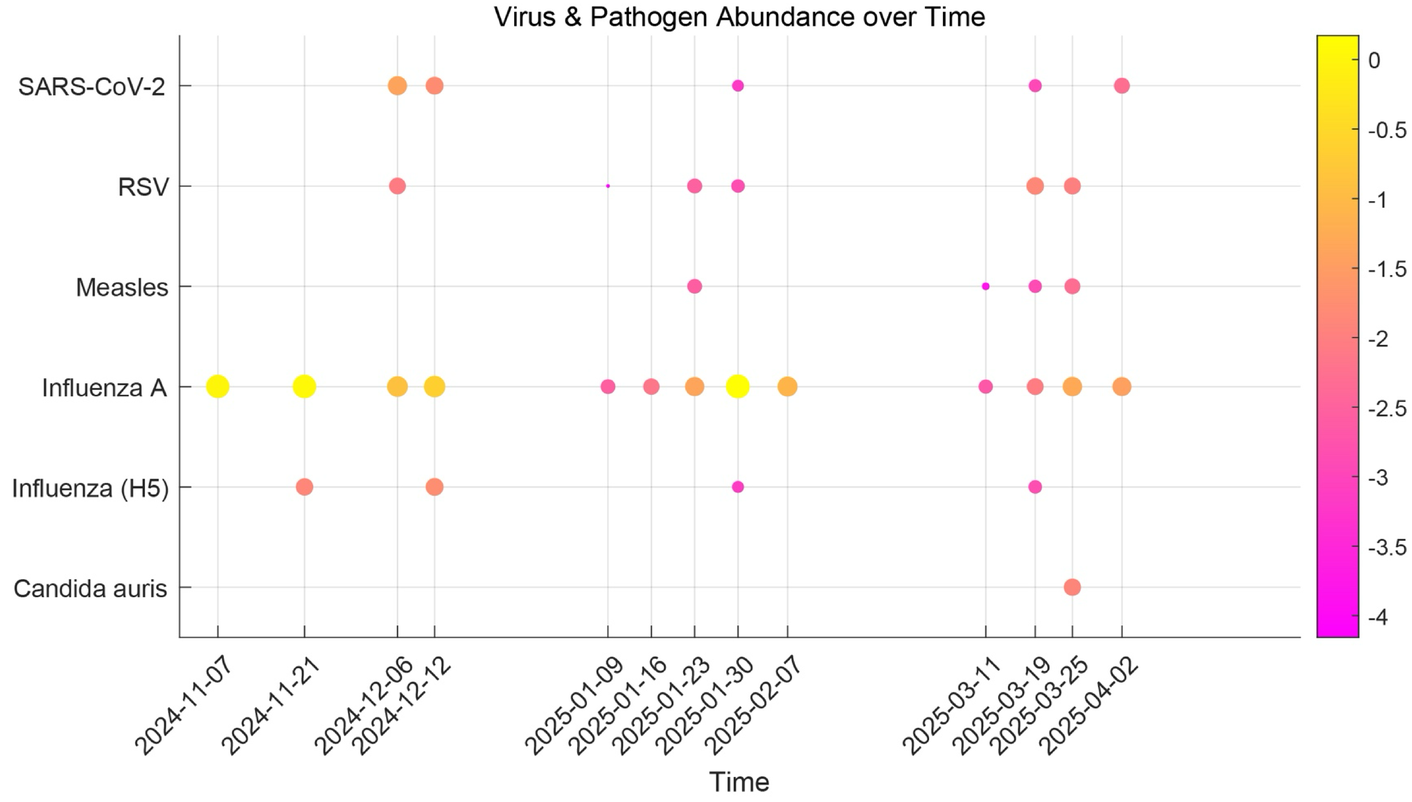




Extra results – Antimicrobial Resistance in WBE



Results



WBE was reflective of classical epidemiology for SARS-CoV-2, RSV and Influenza A.
The detection of Measles and Influenza A subtype H5 despite lack of clinical cases may signify this strategy has a high sensitivity and may yield viral shedding post vaccination (measles) or consumption of dairy products (Influenza A subtype H5).
The detection of Candida auris in wastewater may represent an undetected case/colonized patient in the hospital.

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